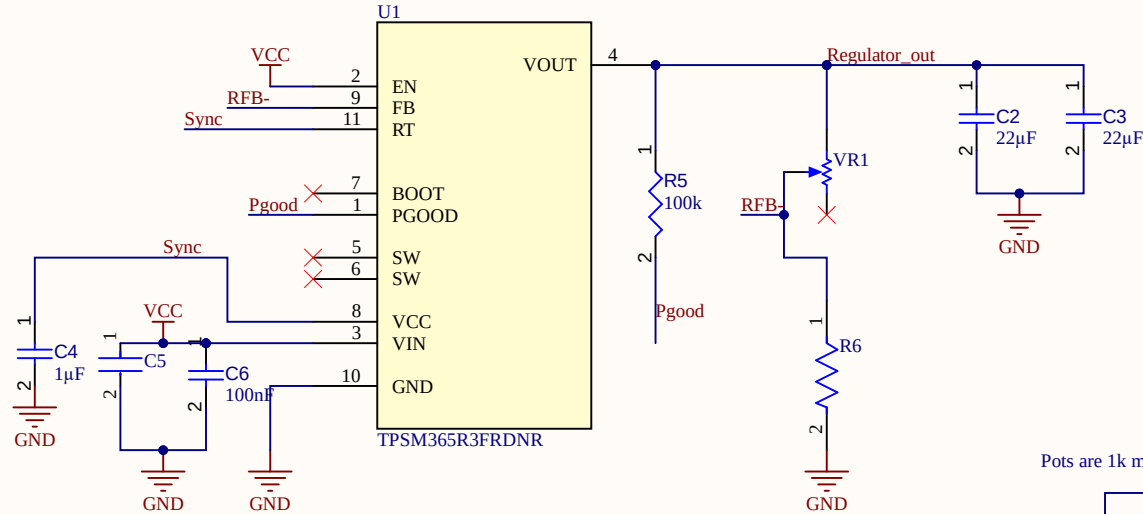
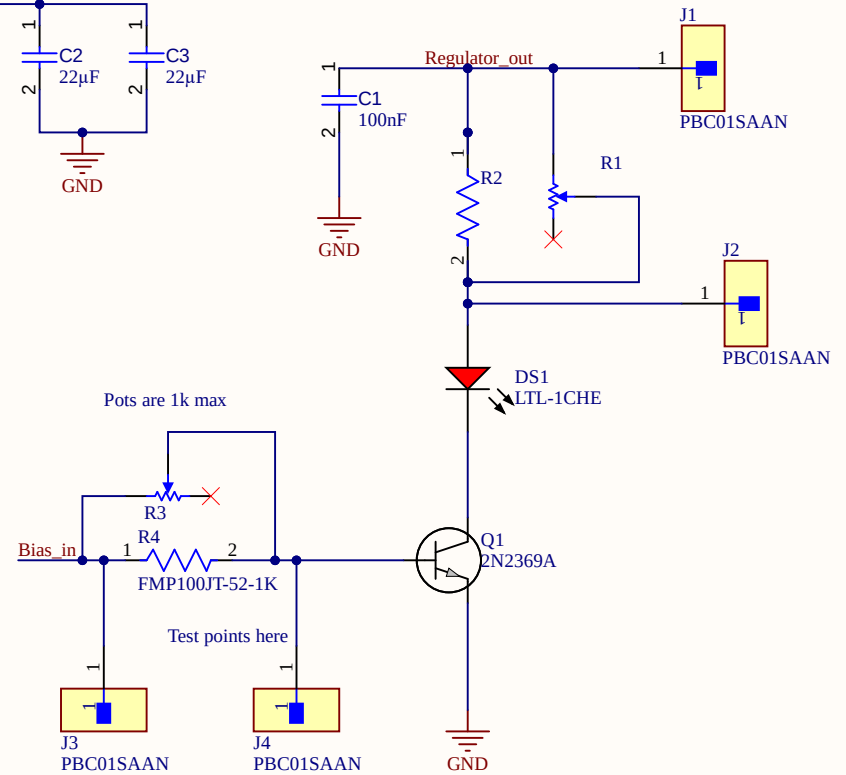


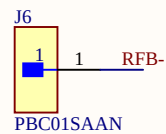
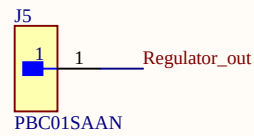
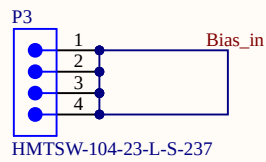
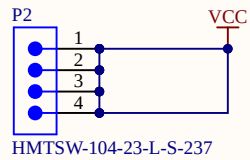
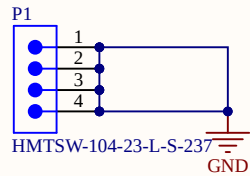
Voltage Regulation



LED Modulation



Connectors

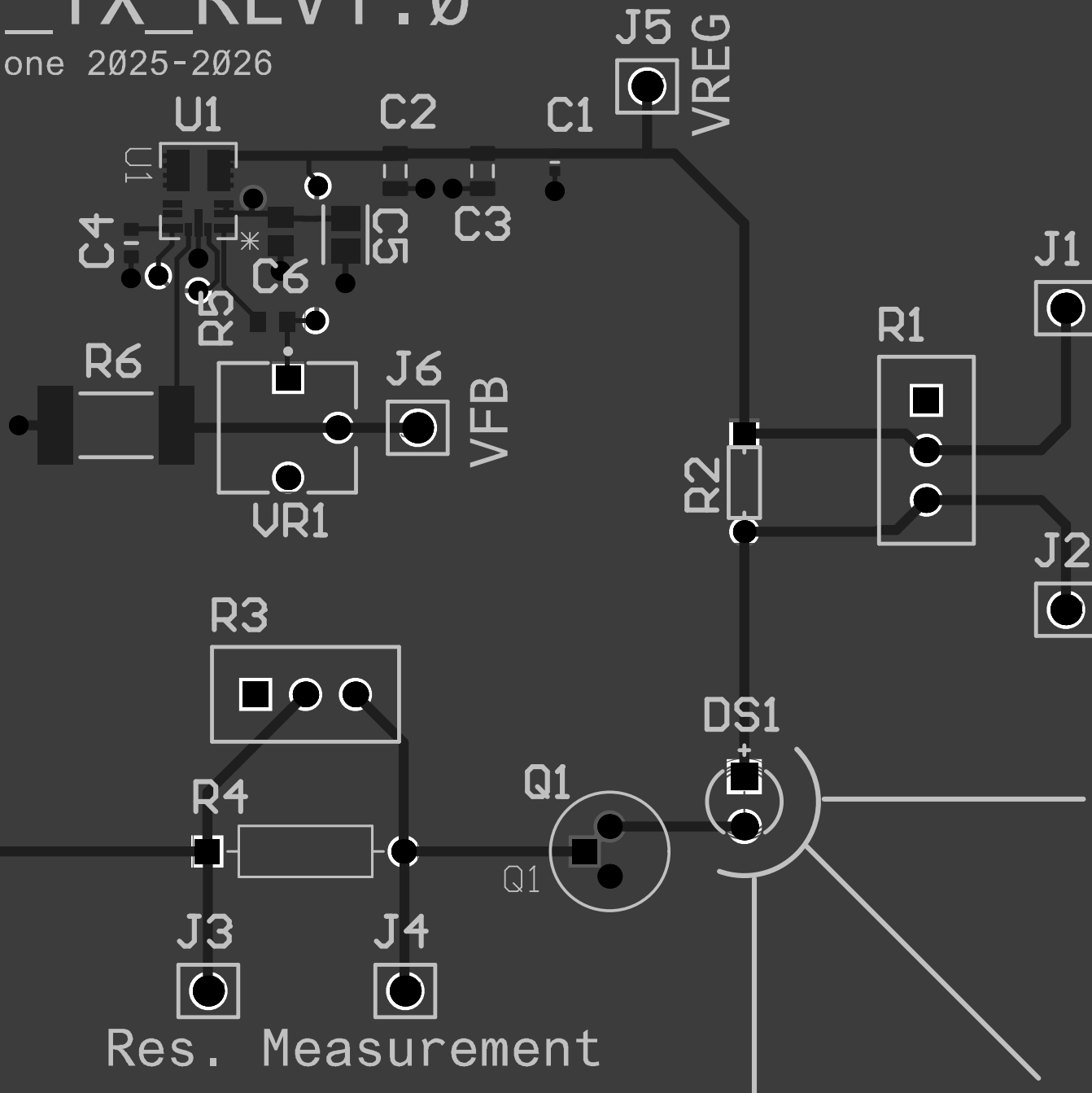
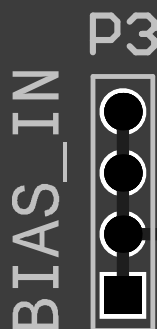
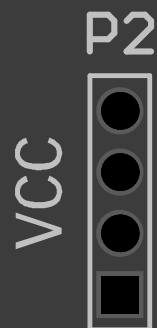


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XCOM_TX_REV1.0

ECE Capstone 2025-2026

P1 GND



Res. Measurement

Res. Measurement

Line #	Name	Description	Designator	Revision ID	Revision Status	Quantity	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
	Capacitor	Chip Capacitor, 100nF +/- 20%, 50V, 0402, Thickness 0.6 mm	C1			1							
	Capacitor	Chip Capacitor, 22uF +/- 40%, 16V, 0805, Thickness 1.45 mm	C2, C3			2							
	Capacitor	Chip Capacitor, 1uF +/- 0.1%, 25V, 0603, Thickness 1 mm	C4			1							
	GRM21B D72A225 MED1K	CAP CER 2.2UF 100V X7T 0805	C5			1							
	Capacitor	Chip Capacitor, 100 nF +/- 10%, 50 V, 0805 (2012 Metric)	C6			1							
	LTL-1CHE	LED Uni-Color Red, 100 mW, 30 mA, -55 to 100 degC, 2-Pin THD, RoHS, Bulk	DS1			1							
	PBC01SA AN	CONN HEADER VERT 1POS	J1, J2, J3, J4, J5, J6			6							
	HIMTSMV-104-23-L-S-237	Modified .025" SQ Post Header, Through-hole, Vertical, -55 to 125 degC, 2.54 mm Pitch, 4-Pin, Male, RoHS	P1, P2, P3			3							
	2N2369A	Description	Q1			1							
	3296W-1-102LF	Square Trimpot(R) Trimming Potentiometer ,1000 Ohm, -55 to 125 degC, 3-Pin THD, RoHS, Tube	RL, R3			2							
	MB40204 0CZ200F C100	RES 220 OHM 0.4W 1% AXIAL	R2			1							
	FMP100J T-52-1K	Resistor, 1 kOhm, ± 5%, 1 W, Axial Leaded, Metal Film, High Power Flame Proof	R4			1							
	Resistor	Chip Resistor, 100 kOhm, +/- 1%, 0.1 W, -55 to 155 degC, 0603 (1608 Metric)	R5			1							
	AC2512F K-0710K	RES SMD 10K OHM 1% 1W 2512	R6			1							
	TPSM365 R3FRDNR	3-V to 65-V input, 1-V to 13 V output, 0.3-A synchronous buck converter power module 11-QFN-FCMOD -40 to 125	U1			1							
	3362P-1-104LF	Square Trimpot(R) Trimming Potentiometer ,100 kOhm, 300 V, -55 to 125 degC, 3-Pin THD, RoHS, Tube	VR1			1							